

Christian S. Kim

chrisk36@engineering.upenn.edu | (516) 350-3850 | [Portfolio Website](#) | github.com/chrisk36

EDUCATION

University of Pennsylvania

B.S.E. Computer Science – Concentration: Computer Vision (GPA: 3.74)

Aug 2024 - May 2028

Philadelphia, PA

EXPERIENCE

Artificial Intelligence Software Engineer Intern

May 2026 – Present

Avionica

Miami, FL

- Accelerated aircraft parameter database generation time from 3–10 days to under 1 hour (>98% reduction) by automating LLM-based extraction, validation, and compliance verification across 89 regulatory flight-data requirements.
- Engineered a multi-stage LLM vision pipeline capable of ingesting heterogeneous aircraft engineering documents (PDF, FRED, TXT, Excel) up to 1,000+ pages, converting OEM documentation from 29+ manufacturers into structured parameter databases.
- Cut document extraction costs by 85% through prompt caching, bounded concurrency, and inference optimization, lowering per-aircraft processing costs from over \$500 to ~\$25.

Web Redesign Team Lead

Jan 2026 – May 2026

Atelic Travel Agency / Penn SPARK Client Team

Philadelphia, PA

- Led a 10-person cross-functional team of designers and engineers through the end-to-end redesign of a client-facing travel platform, managing 5 product milestones across weekly Agile sprints and delivering the project ahead of schedule.
- Served as the sole client-facing technical lead, leading weekly stakeholder meetings and translating 20+ business requirements into engineering specifications that guided frontend development and improved usability and brand consistency.

Soft Robotics Research Engineer

May 2025 – Jul 2025

National University of Singapore

Singapore, SG

- Designed and validated a light-responsive soft robotic platform that converted optical stimulation into controlled mechanical motion, integrating hydrogel substrates with electronic hardware to demonstrate repeatable real-time actuation.
- Developed a biodegradable light-emitting hydrogel with 50% higher luminescence and 40% greater mechanical stability through iterative material optimization, fabrication, and experimental performance testing.

PROJECTS

SudoBuddy | TensorFlow • TensorFlow Lite Micro • React

Nov 2025 – Dec 2025

Machine Learning Engineer / Personal Project

Philadelphia, PA

- Optimized a computer vision pipeline for puzzle localization, cell segmentation, and handwritten digit recognition, enabling 95–99% accuracy with sub-100 ms inference despite the limited memory and compute resources of an Arduino XIAO microcontroller.
- Quantized and deployed a custom CNN using TensorFlow Lite Micro, reducing model size by 4× while preserving 95–99% accuracy and enabling real-time embedded inference within strict memory and compute constraints.
- Integrated computer vision, voice recognition, and LED visual feedback into a fully embedded hands-free solving system, eliminating the need for external displays or manual input while enabling an intuitive real-time user experience.

Roomies | React Native • Tailwind CSS

Jan 2025 – May 2025

Frontend Developer / Penn SPARK

Philadelphia, PA

- Drove a 30% increase in daily task completion across 50+ beta users by implementing a push notification system and points-based reward model, strengthening user engagement during pre-launch testing within an 8-person cross-functional team.
- Directed frontend development of a cross-platform React Native application, architecting task assignment workflows, a personalized dashboard, and a gamified leaderboard using React Native and Tailwind CSS.

LEADERSHIP

Global Programs Leader

May 2025 – Present

International Student & Scholar Services, University of Pennsylvania

Philadelphia, PA

- Co-led the planning and execution of a 3-day pre-orientation retreat for 40 international first-year students, designing team-building activities that fostered intercultural engagement, meaningful peer connections, and a smooth transition to campus life.
- Mentored students through 1-on-1 guidance and monthly structured programming, supporting academic, cultural, and social transition.

Marketing Strategy Consultant – Vice President

Sep 2024 – May 2026

Penn Marketing Undergraduate Student Establishment

Philadelphia, PA

- Led a 7-person consulting team to 3rd-place finishes among 90+ teams at the 2025 & 2026 AMA International Collegiate Case Competitions.
- Developed product and marketing strategies for MLB and Adobe Express by designing AI-driven features, UX improvements, and Gen Z engagement initiatives projected to increase user engagement by 60% and 40%, respectively.

SKILLS & INTERESTS

Languages: Python, Java, TypeScript/JavaScript, SQL, C/C++, OCaml

AI & Frameworks: TensorFlow, TensorFlow Lite Micro, PyTorch, OpenCV, Hugging Face, React, React Native, Node.js, Express.js, Tailwind CSS

Tools & Platforms: Docker, Git, Azure, AWS, PostgreSQL, MongoDB, Firebase, REST APIs, Figma, Autodesk Maya

Spoken Languages: English, Korean (Fluent), Spanish (Fluent)

Interests: Dance, Video Game Development, Marketing, Volleyball, Soccer, Swimming